

Virtual embodiment reduces fear of movement after knee surgery

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ABSTRACT

Virtual embodiment and virtual exercising are a promising adjunct to treatment for orthopedic patients, who face challenges from physical and psychological barriers during rehabilitation. This randomized controlled trial evaluated the impact of VR on post-surgical knee recovery. Patients ($n=47$) were assigned to either conventional rehabilitation or conventional plus daily 20-minute VR training, which utilized embodied avatars for movement visualization. The VR group demonstrated a substantial reduction in kinesiophobia scores at 4 weeks post-surgery ($p=0.043$). No significant differences were observed in quadriceps strength, knee range of motion, and disability scores. These findings support VR's potential for addressing psychological barriers in rehabilitation.

Index terms: Immersive virtual reality; embodiment; rehabilitation; orthopedics; traumatology.

1 INTRODUCTION

Immersive virtual reality (VR) offers a unique platform for integrating motor imagery, action planning, and action observation using virtual embodiment [5]. By creating the illusion of owning a virtual body, patients can simultaneously imagine and observe movements of a virtual body that they perceive as their own [4]. This can potentially accelerate orthopedic rehabilitation by restoring normal sensorimotor processing, since altered visual feedback from a moving virtual limb can also alter real limb range motion and muscle activity [1]. Observing a healthy virtual limb moving normally and performing challenging exercises may reduce fear associated with movement and exercise, contributing to a graded exposure approach for managing fear-avoidance behavior [2].

The study aimed to investigate if home-based VR rehabilitation, added to standard care, improves functional and psychological outcomes for patients recovering from anterior cruciate ligament (ACL) reconstruction. The researchers hypothesized that embodying a virtual body and experiencing it moving and exercising would lead to decreased disability, reduced kinesiophobia, increased strength, and improved range of motion compared

to controls at 4 weeks post-surgery, with these benefits persisting at 3 and 9 months post-surgery.

2 MATERIALS AND METHODS

2.1 Study Design and Participants

A parallel randomized controlled trial was conducted, involving two groups of patients ($n=47$). The study was pre-registered with the US National Agency of Medicine (NCT05080894).

Patients aged 18-65 undergoing ACL reconstruction were recruited from a large public teaching hospital. Exclusion criteria included previous knee surgery, prior ACL rupture, tibiofemoral arthritis, epilepsy, severe visual impairments, and pregnancy.

2.2 Randomization and Blinding

Patients were randomly allocated to either an experimental or a control group on the days of surgery using randomization software.

2.3 Interventions

Control group: Patients had conventional treatment, including: educational video, ergonomic advice for daily activities, light range of motion exercises, isometric lower limb strengthening, and partial weight-bearing until formal physiotherapy started (4 weeks post-surgery).

VR treatment group: In addition to conventional treatment, patients engaged in a home-based VR rehabilitation program for 4 weeks, 5 days a week, for a minimum of 20 minutes daily.

The VR program involved embodying a gender-matched virtual avatar in a relaxing virtual meadow environment. Visuomotor congruence between the patient's upper limb and head movements and the avatar's movements, along with visuotactile stimulation, enhance the sense of embodiment.

First the user observes the exercise (third-person perspective, action observation), followed by mentally visualizing and planning the movement, before initiating the exercise (motor imagery) using the controller (providing a sense of agency). The patient does not perform the exercise physically but observes their embodied virtual legs performing it. The exercises consisted of hip, knee, and ankle exercises and games that progressively increase in complexity and speed.

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2.4 VR equipment

The VR system used was an Oculus Quest (Meta, CA, USA). The program was developed in Unity 2019.3.

2.5 Outcome measures

The primary outcome measure was self-reported disability assessed using the IKDC Questionnaire. Secondary outcome measures included quadriceps strength, knee range of motion, and kinesiophobia (assessed using the Tampa Scale of Kinesiophobia-short form (TSK-11)). Assessments were conducted at baseline (day 0-1 post-surgery), 4 weeks (post-treatment), 12 weeks, and 36 weeks.

2.6 Statistical analysis

Statistical analysis was performed using JASP 0.18.2. Demographic and clinical data were compared between groups using chi-square tests or *t*-tests. RM-ANOVA was used to analyze the outcome measures.

2.7 Ethical considerations

Ethical approval was obtained from the ethics committee at the Hospital Clinic de Barcelona and the study followed the Declaration of Helsinki.

3 RESULTS

3.1 Recruitment/baseline characteristics

Forty-seven patients were recruited, with 44 completing the study (21 in the VR treatment group and 23 in the control group). Three patients dropped out due to personal reasons or reinjury. No adverse effects were reported, although two patients had mild dizziness that resolved with continued VR use. No differences in demographic or baseline characteristics were observed between the groups.

3.2 Outcomes

A significant between-group difference was observed (RM-ANOVA, $p=0.048$). Post-hoc analysis revealed a significant difference at 4 weeks ($p=0.043$, Tukey's HSD), favoring the VR treatment group, with a mean reduction in TSK-11 score of 4.8 points. No between-group differences were found in disability, range of motion or quadriceps score at any time point.

4 DISCUSSION

The VR intervention demonstrated a positive effect on reducing kinesiophobia, particularly immediately after treatment at 4 weeks. This finding shows the importance of addressing kinesiophobia in ACL rehabilitation, as it can hinder recovery and increase reinjury risk. Early intervention for kinesiophobia is recommended in clinical practice guidelines [3]. The VR program, delivered during the initial 4 weeks when patients are typically self-managing at home before formal rehabilitation, may be beneficial in this early post-operative phase.

The positive impact of VR on kinesiophobia is

consistent with previous research [6], although most high-quality studies have focused on low back pain and non-immersive VR interventions. This study contributes to the limited literature on the effects of VR on kinesiophobia in lower limb conditions. The study's findings regarding self-perceived disability, strength, and range of motion contrast with a previous study, where a similar VR intervention showed positive effects on these outcomes in patients with upper limb fractures [4]. This discrepancy may be attributed differences in levels of immobilization, adherence to treatment, as well as type and location of injury, differences in injury type and location.

Limitations of the study include lack of blinding of patients and assessors, heterogeneity in the study population regarding demographics and injury/surgery type, and the absence of return-to-sport time as an outcome measure due to the complexity and heterogeneity of these decisions.

Using VR with an embodied virtual body during the early stages of ACL rehabilitation is a feasible approach to reduce kinesiophobia, a significant barrier to recovery and a risk factor for reinjury. However, the VR intervention, just providing visual feedback, did not impact strength or disability in this patient group. Future research should investigate the effects of embodiment in other orthopedic conditions and stages of recovery to gain a better understanding of its efficacy. Exploring different VR interventions, such as incorporating more active exercises, gamification elements, and greater personalization, may enhance adherence and potentially accelerate recovery.

5 ACKNOWLEDGEMENTS

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